Annexure 02

Teaching & Examination Scheme and Syllabus

B. Tech Electrical Engineering

3rd and 4th Semester

Batch – EE2020

w.e.f. ACY 2021-2022

Approved by BoS: March 2021

PANDIT DEENDAYAL ENERGY UNIVERSITY, GANDHINAGAR
SCHOOL OF TECHNOLOGY

COURSE STRUCTURE FOR B. TECH. IN ELECTRICAL ENGINEERING

Semester IV			B. Tech. in Electrical Engineering										
Sr. No.	Course/Lab Code	Course/Lab Name	Teaching Scheme					Examination Scheme					
								Theory			Practical		Total
			L	T	P	C	Hrs/Wk	MS	ES	IA	LW	LE/Viva	Marks
12.	20EE205T	Electric Power Generation, Transmission and Distribution	4	0	0	4	4	25	50	25	--	--	100
13.	20EE206T	Linear Control Theory	3	0	0	3	3	25	50	25	--	--	100
14.	20EE206P	Control Systems Laboratory	0	0	2	1	2	--	--	--	50	50	100
15.	20EE207T	Synchronous and DC Machines	3	0	0	3	3	25	50	25	--	--	100
16.	20EE207P	Synchronous and DC Machines Laboratory	0	0	2	1	2	--	--	--	50	50	100
17.	20EE208T	Microprocessors and Microcontrollers	3	0	0	3	3	25	50	25	--	--	100
18.	20EE208P	Microprocessors and Microcontrollers Laboratory	0	0	2	1	2	--	--	--	50	50	100
19.	20EE209T	Open Elective II	3	0	0	3	3	25	50	25	--	--	100
20.	20IF201T	Industry 4.0	2	0	0	2	2	25	50	25	--	--	100
21.	20IF201P	Industry 4.0 Laboratory	0	0	2	1	2	--	--	--	50	50	100
22.	TP210	Industrial Orientation	0	0	0	1	0	--	--	--	--	100	100
		Total	**18**	**0**	**08**	**23**	**26**						**1100**

IA- Internal Assessment, MS-Mid Semester; ES – End Semester Exam

Department will offer Open Elective based on availability of expertise/faculty. This open elective will be offered to the students enrolled in B.Tech. programs of School of Technology other than Electrical Engineering.

Student of Electrical Engineering will have a choice for one open elective course from the basket of open electives announced by the other departments of School of Technology/other schools of university.

Subject Code	Open Elective II
20EE209T	Energy and Water Nexus

20EE205T					Electrical Power Generation, Transmission and Distribution					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
4	0	0	4	4	25	50	25	--	--	100

COURSE OBJECTIVES

1. To understand the electrical power generation and its economics
2. To calculate parameters of transmission line and analyse its performance
3. To introduce design aspects of transmission line.
4. To understand characteristic of distribution systems

UNIT I: ELECTRICAL POWER GENERATION AND LOADS **12 Hrs**

Introduction to structure of power system. Schematic diagram for thermal, gas, hydro and nuclear power plant and associated equipment and auxiliaries. Load curves and load duration curves, associate terms and factors, importance of high load factor, different types of loads, tariffs, power factor improvement methods, derivation of most economical power factor,

UNIT II: OVERHEAD TRANSMISSION LINE **14 Hrs**

Types of conductors, resistance of transmission line. Calculation of inductance and capacitance of single phase & three phase line with different line configuration, Concepts of GMD, GMR, transposition of line. Characteristic and performance analysis of short, medium and long lines through ABCD parameters, voltage regulation and transmission efficiency of lines under different loading conditions, Surge Impedance Loading, Ferranti effect, active and reactive power flow through tine and their impact on bus voltage magnitude and phase angle, shunt and series compensation of line.

UNIT III: ELECTRICAL AND MECHANICAL DESIGN OF TRANSMISSION LINE **12 Hrs**

Mechanical Design of Overhead Lines: Choice of line insulators, voltage distribution across insulators disc, string efficiency, methods of equalizing potential across insulator. Sag and tension calculations, Effect of ice and wind, Stringing chart, Sag template, Tower design, Spacing and clearance, Vibration damper **Electrical Design of Line:** Choice of voltage level, kW-km loading, choice of conductors, their spacing and current carrying capacity, consideration of corona discharge and corona loss. Sample case study for complete design of transmission line.

UNIT IV: DESIGN OF DISTRIBUTION SYSTEM AND SUB STATIONS **14 Hrs**

Radial, loop and grid type sub-transmission, distribution substation bus schemes, location and rating of substation, AC power supply scheme, advantage of higher transmission voltage, various supply schemes in distribution systems, voltage drop and power loss calculations in distribution systems, types of cables, economics of power supply considering economic choice of conductor size and voltage level. **Power station and substation grounding:** Objectives of grounding, definitions, tolerable limits of body currents, soil resistivity, measurement of soil resistivity, earth resistance, measurement of earth resistance, tolerable step and touch potential, actual step and touch potential, design of earthing grid.

TOTAL HOURS **52 Hrs**

COURSE OUTCOMES

On completion of the course, student will be able to

- CO1 – Understand basic schematic of power plant based on conventional energy sources and understand different terms to study economics of power generation
- CO2 – Derive transmission line parameters with different configurations of transmission line
- CO3 – Analyse the performance of short, medium and long transmission lines and compute its voltage regulation, power transfer capability, efficiency, compensation methods
- CO4 – Perform electrical and mechanical design of transmission line
- CO5 – Understand design of distribution supply systems
- CO6 – Understand the design concepts of substations and its grounding practices

TEXT/REFERENCE BOOKS

1. Sivanagaraju and Satyanarayana, **"*Electrical Power Transmission and Distribution*", Pearson Education**
2. Glover, Sarma, Overbye, "***Power System Analysis and Design***", Cengage Publication.
3. M.V. Deshpande, "***Electrical Power Stations***", PHI Publications
4. V. K. Mehta, Rohit Mehta, ***"Principles of Power System",*** S Chand Publications.
5. Dr. S.L. Uppal and S. Rao, **"*Electrical Power Systems*", Khanna Publishers.**
6. Kothari &Nagrath***, "Power System Engineering,"*** Tata McGraw-Hill Education, 2008**.**
7. Hadi Saadat, **"Power System Analysis",**Tata McGraw-Hill

END SEMESTER EXAMINATION QUESTION PAPER PATTERN

Max. Marks: 100	**Exam Duration: 3 Hrs**
Part A/Question: 4 Questions, one from each unit, each carrying 15 marks	60 Marks
Part B/Question: 4 Questions, one from each unit, each carrying 10 marks	40 Marks

20EE206T					Linear Control Theory					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
3	0	0	3	3	25	50	25	--	--	100

COURSE OBJECTIVES

1. To teach the basic concepts of control theory and feedback systems.
2. To start thinking on the concepts of mathematical modelling and to model simple physical systems.
3. Introduce to concept of state, state space and state space models of a system
4. To teach students the characteristics of closed-loop control systems, including steady-state and transient response.
5. To understand and analyze the concepts of stability through time and frequency domain methods.

UNIT I: INTRODUCTION TO CONTROL SYSTEM **5 Hrs**

Definition, Open loop and closed loop systems, Examples, Components of control systems, Types of control systems, Concept of feedback, positive and negative feedback. Types of processes and its characteristics, linear, non-linear, time-varying, time-invariant, continuous, discrete process, lumped parameter, distributed parameter processes.

UNIT II: MATHEMATICAL MODELLING OF PHYSICAL SYSTEMS **15 Hrs**

First principle modelling, Modelling of physical systems such as electrical, mechanical, electro-mechanical systems, analogous systems, concept of transfer function, poles, zeroes, 'order' and 'type' of the system, computation of overall transfer function, block diagram reduction techniques, signal flow graphs, Introduction to state, state space and state variables, state space models, conversion of state space models to transfer function models, solution of state equation, Homogeneous and Non-homogeneous state equations, State transition matrix, concept of controllability and observability.

UNIT III: TIME RESPONSE ANALYSIS **05 Hrs**

System dynamics, Standard test signals, Step, Impulse, Ramp, Parabolic, Sinusoidal signals, Transient and steady state response of first and second order systems, Time response specifications, Steady state error, Concept of dominant poles for higher order systems.

UNIT IV: STABILITY ANALYSIS OF CONTROL SYSTEMS **15 Hrs**

Notations of stability, Asymptotic and BIBO stability, Necessary conditions for stability, Routh-Hurwitz stability criterion, Relative stability, Introduction to root locus, definition, rules to construct root locus, Stability analysis and control system design using root locus, Introduction to frequency response, Frequency response specifications, Stability analysis using Bode plots, Polar and Nyquist plots.

TOTAL HOURS 40 Hrs

COURSE OUTCOMES

On completion of the course, student will be able to:

CO1 – Understand the basic concepts of control theory and feedback control systems.
CO2 – Apply the first principle laws to model (Mathematically) the simple physical processes.
CO3 – Analyze the time response of continuous systems.
CO4 – Evaluate the stability of control systems using Routh array test and Evans root locus technique.
CO5 – Analyze the performance characteristics and stability of a control system using frequency response methods.
CO6 – Understand the concept of state, state variables and state space modelling

TEXT/REFERENCE BOOKS

1. I.J. Nagrath and M.Gopal, **Control system Engineering**, New Age International Limited.
2. Katsuhiko Ogata, Modern **Control Engineering**, PHI Learning Pvt. Ltd., New Delhi.
3. Gene F. Frankline, J. David Powell, Abbas Emami-Naeini, **Feedback Control of Dynamic Systems**, Pearson Education Inc.
4. I.J. Nagrath and M. Gopal,**Systems Modelling and Analysis**, Tata McGraw-Hill Publishing Company Limited.
5. Norman N. Nise, **Control system engineering**, Wiley International Edition.

END SEMESTER EXAMINATION QUESTION PAPER PATTERN

Max. Marks: 100	**Exam Duration: 3 Hrs**
Part A/Question: 4 Questions, one from each unit, each carrying 10 marks	40 Marks
Part B/Question: 4 Questions, 2 from unit II and 2 from unit IV, each carrying 15marks	60 Marks

20EE206P					Control Systems Laboratory					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
0	0	2	1	2	--	--	--	50	50	100

COURSE OBJECTIVES

1. To make the students acquainted with the use of computational software such as MATLAB and its programming.
2. Students will able to perform simulation exercise on control theory using Simulink environment.
3. Students will be able to identify model parameters using open loop experimental runs.
4. To study stability analysis using time and frequency response methods using MATLAB.
5. To do transient and steady state response analysis of processes for standard test signals.

LIST OF EXPERIMENTS

1. Introduction to MATLAB (Mathematical Computational & Simulation Software) - Applications in Control Theory.
2. To study mathematical modelling of Electrical Systems & Simulate using MATLAB.
3. To study mathematical modelling of Mechanical Systems & Simulate using MATLAB.
4. Transient response analysis of first order systems (Simulation Exercise using MATLAB).
5. Transient response analysis of second order systems (Simulation Exercise using MATLAB).
6. To solve ordinary differential equations (state space models) using solvers in MATLAB.
7. To solve ordinary differential equations (state space models) using Simulink in MATLAB.
8. To study transient and steady state response of liquid level system using open loop experimental runs and verify them using open loop simulations.
9. To find time constant and steady state gain of Single Board Heater System (SBHS).
10. To study stability analysis using Root Locus (verifying using MATLAB).
11. To study stability analysis using Bode Plots (verifying using MATLAB).
12. To check controllability and observability of the system using state space models (verifying using MATLAB).

COURSE OUTCOMES

On completion of the course, student will be able to

CO1 – Understand the use of various commands and functions used in control theory using MATLAB Computational Software.
CO2 – Analyze the transfer function and differential equation models (state space models) using MATLAB simulation environment.
CO3 – Perform hands-on experiments to analyze the dynamics of the system.
CO4 – Evaluate the stability of a system using Root Locus and Bode plots in MATLAB environment.
CO5 – Analyze the time response of continuous systems using MATLAB simulation environment.
CO6 – Model the process using time response data.

END SEMESTER LABORATORY EXAMINATION PATTERN

Max. Marks: 100

Continuous evaluation	50 marks
End semester examination and Viva-voce	50 marks

20EE207T					Synchronous and DC Machines					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
3	0	0	3	3	25	50	25	--	--	100

COURSE OBJECTIVES

1. To understand basic principles, working, construction, classification and operating characteristics of rotating machines such as Synchronous and DC machines
2. To evaluate and analyse the performance of Synchronous and DC machines
3. To enable students to identify and solve the related problems in Synchronous and DC machines
4. To impart sound knowledge about the different applications of Synchronous and DC machines

UNIT I: THREE PHASE SYNCHRONOUS GENERATOR **18 Hrs**

Construction, Types, Operating Principle, Terms Related to AC Armature Winding, Pitch & Distribution Factors, Generated EMF, Effect of Harmonics on Induced EMF, Operation on Load – Standalone operation of an Alternator, Phasor Diagrams, Power Output Equation, Armature Reaction, SCR of an Alternator, Voltage Regulation, Determination of Voltage Regulation by Direct Loading, Synchronous Impedance, MMF & ZPF Methods, Two Reaction Theory, Condition of Parallel Operation, Synchronizing of Alternators & Methods of Synchronization, House Diagram, Operation of two or more machines, load sharing, Operation on Infinite Bus, F-P, Q-V Characteristics, Effect of change in excitation, Effect of change in prime mover speed, Slip Test, Hunting of Synchronous Machines & Its Prevention, Capability Curves, Short Circuit Transients, Short Circuit on Generator, Constant Flux Linkage Theorem, Mathematical Solution of series RL circuit, Transient, Sub-transient, Steady state
Single Phase Generator Short Circuit, Testing and Trouble Shooting of Three Phase Alternator

UNIT II: THREE PHASE SYNCHRONOUS MOTOR **06 Hrs**

Construction, Operating Principle, Phasor Diagrams, Starting Methods, Effect of Varying Field Current at Different Loads, V-Curves, Hunting & Damping, Synchronous Condenser, Power Developed by Synchronous Motor.

UNIT III: DC GENERATOR **10 Hrs**

Principle & Construction of a DC Machine, Types of DC Generators, DC Machine Armature Winding, Characteristics of DC Generators, EMF Equation, Voltage Build-Up in a Shunt Generator, Critical Resistance & Speed, Losses in DC Machine, Power Stages & Efficiency, Armature Reaction & its Effects, Inter Poles & Compensating Winding, Commutation & Methods to Improve Commutation, Applications.

UNIT IV: DC MOTOR **06 Hrs**

Working Principle, Back EMF, Voltage & Power Equations, Types, and Torque of a DC Motor, Power Stages, Efficiency, Performance Characteristics, Necessity of Starter, Three & Four Point Starters, Speed Control of DC Motors, Applications, **Efficiency and Testing of DC Machines:** Methods of Testing, Brake Test, Swinburne's Test, Hopkinson's Test, Field Test, Retardation Test, and Trouble Shooting of DC Machines.

TOTAL HOURS 40 Hrs

COURSE OUTCOMES

On completion of the course, student will be able to:

CO1 – Understand the construction, principle of operation and working of different kind of rotating machines
CO2 – Understand operation and performance of three phase alternator in isolated & grid connected mode
CO3 – Calculate the voltage regulation of alternator by different methods
CO4 – Appraise the purpose for parallel operation of generators and learn the process of synchronization
CO5 – Analyze the performance of DC machines by various tests
CO6 – Compare the different methods of starting and speed control of DC motors

TEXT/REFERENCE BOOKS

1. B.L. Theraja & A.K. Theraja, **A Text Book of Electrical Technology Volume II-AC & DC Machines**, S. Chand & Co.
2. J.B.Gupta, **Theory and performance of electrical machines**, Katson Publication, S.K.Kataria& Sons.
3. Ashfaq Hussain, **Electric Machines**, DhanpatRaiand Co.
4. P. S. Bimbhra, **Electrical Machinery**, Khanna Publishers
5. M. G. Say, **Alternating Current Machines**, Pitman and Sons
6. J. G. Jamnani, **Electrical Machines**, Mahajan Publishing House.

END SEMESTER EXAMINATION QUESTION PAPER PATTERN

Max. Marks: 100	**Exam Duration: 3 Hrs**
Part A/Question: 4 Questions, one from each unit, each carrying 15 marks	60 Marks
Part B/Question: 4 Questions, 3 form unit I and 1 form unit III, each carrying 10 marks	40 Marks

20EE207P					Synchronous and DC Machines Laboratory					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
0	0	2	1	2	--	--	--	50	50	100

COURSE OBJECTIVES

1. To impart the practical knowledge of construction and working of rotating electrical machines
2. To determine the performance characteristics of rotating electrical machines by different tests
3. To determine the voltage regulation of alternator by different methods
4. To learn the synchronization procedure for alternators

List of Experiments:

1. To determine the voltage regulation of an alternator by direct loading method.
2. To determine the voltage regulation of an alternator by synchronous impedance method.
3. To determine the voltage regulation of an alternator by MMF method.
4. To determine the voltage regulation of an alternator by Z.P.F. method.
5. Synchronization of two 3-phase alternators with common bus bars by various methods.
6. To plot `V' curve of three phase synchronous motor.
7. To determine direct and quadrature axis reactance of a salient pole alternator by slip test.
8. To understand the construction and operation of DC machine by working cut section.
9. To obtain the efficiency and load characteristics of a DC shunt motor by direct load test.
10. To determine external and internal characteristics of a DC shunt generator.
11. To determine external and internal characteristics of a DC series generator.
12. To determine external and internal characteristics of a DC compound generator.
13. To control speed of a DC shunt motor by armature control and field control methods.
14. To find the efficiency of a DC machine by Swinburne's test.
15. To perform Hopkinson's Test on a pair of two identical DC machines to find the efficiency of each machine.
16. To perform Field test on a pair of two identical DC series machines to find the efficiency

COURSE OUTCOMES

On completion of the course, student will be able to:

CO1 – Demonstrate construction, working and operating of rotating machines such as Synchronous and DC machines
CO2 – Analyze the performance parameters and characteristics of DC machines
CO3 – Measure and compare the voltage regulation of Alternator by different method
CO4 – Understand the speed control of DC motors
CO5 – Understand the parallel operation of Alternators
CO6 – Verify and Analyze the performance characteristics of rotating machines by conducting different tests.

END SEMESTER LABORATORY EXAMINATION PATTERN

Max. Marks: 100

Continuous evaluation	50 marks
End semester examination and Viva-voce	50 marks

20EE208T					Microprocessors and Microcontrollers					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
3	0	0	3	3	25	50	25	--	--	100

COURSE OBJECTIVES

1. To get familiar with microprocessors and microcontrollers and their role in designing embedded systems.
2. To understand the architecture, assembly programming and timing diagram for a microprocessor and microcontroller
3. To write basic programs in C to explore the functionalities of microcontrollers for real-world applications

UNIT I:FUNDAMENTALS OF MICROPROCESSORS **10 Hrs**

Introduction to Microprocessors:8-bit microprocessor and microcontroller architecture, Comparison of 8-bit, 16-bit and 32-bit microcontrollers, Definition, classification and examples of Embedded systems

Internal architecture of Intel 8085 microprocessor: Block diagram, Registers, Internal Bus Organization, Functional details of pins, Registers, ALU

UNIT II:8085 MICROPROCESSOR **10 Hrs**

Memory Interfacing: Interfacing external RAM and ROM, Bus System, Control signals, Address / Data bus multiplexing and de-multiplexing.

Assembly language programming: 8085 instruction set: Instructions, Classifications, Addressing modes, Decision Making, Looping, Stack and Subroutines etc. and Programming examples. Timing Diagrams of various instructions, Interrupts.

UNIT III:8051 MICROCONTROLLER **10 Hrs**

Introduction to 8051 microcontroller: Introduction, Architecture of 8051 Microcontroller, 8051 microcontroller hardware, Pin diagram of 8051, input/output pins, ports and circuits. Internal RAM and ROM, SFR's

On-board Peripherals: GPIO, Timers and Counters, Interrupt, Serial data communication (UART)

UNIT IV:8051 C PROGRAMMING **10 Hrs**

Basics of C programming, programming examples for GPIO, Timers, Round-robin with interrupts, PWM, LCD, UART

TOTAL HOURS 40 Hrs

COURSE OUTCOMES

On completion of the course, student will be able to:

CO1 – Understand microprocessor & microcontroller and their applications
CO2 – Learn the architecture, assembly language programming and timing diagram for a microprocessor & microcontroller
CO3 – Understand structure of embedded C programming
CO4 – Explore the functionalities of microcontrollers for real-time applications
CO5 – Illustrate the functionalities of different peripherals and their interfacing with microprocessor
CO6 – Develop the embedded solution for real time applications

TEXT/REFERENCE BOOKS

1. Ramesh S. Gaonkar, "**Microprocessor Architecture, Programming, and Applications with the 8085**", Penram International.
2. David E. Simon, "**An Embedded Software Primer**", Addison-Wesley Professional
3. William Kleitz, "**Microprocessor and Microcontroller fundamentals: The 8085 and 8051 Hardware and Software**"
4. Douglas V. Hall , "**Microprocessors, Interfacing and Peripherals**", Tata McGraw Hill,
5. Ajoy Ray, K Bhurchandi, "**Advanced Microprocessors and Peripherals**", Tata McGraw Hill
6. Muhammad Ali Mazidi, Janice GillispieMazidi and RolinMcKinley, "**The 8051 Microcontroller and Embedded Systems Using Assembly and C**", Pearson Education.
7. K. J Ayala, D. V. Gadre, "**The 8051 Microcontroller and Embedded Systems using Assembly and C**", Cengage Learning, India Edition.

END SEMESTER EXAMINATION QUESTION PAPER PATTERN

Max. Marks: 100 **Exam Duration: 3 Hrs**

Part A/Question: 4 Questions, one from each unit, each carrying 15 marks 60 Marks
Part B/Question: 4 Questions, one from each unit, each carrying 10 marks 40 Marks

20EE208P					Microprocessors and Microcontrollers Laboratory					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
0	0	2	1	2	--	--	--	50	50	100

COURSE OBJECTIVES

1. To introduce 8085 architecture and programming assembly language.
2. To introduce basic concepts of interfacing memory and peripheral devices to microprocessor.
3. To introduce 8051 microcontroller.
4. To write basic programs in C to explore the functionalities of microcontrollers for real-world applications

List of Experiments:

1. Tabulate Instruction Set for 8085 microprocessors
2. Write a program to move a data block starting at Memory Location X to Location Y without overlap. X, Y and block length are to be specified
3. Write a program to move a data block starting at Memory Location X to Location Y with overlap. X, Y and block length are to be specified
4. Write a program to perform an addition of two 8 bit numbers stored at locations X and Y.
5. Write a program to perform subtraction of two 8 bit numbers stored at location X and Y.
6. Find the product of two unsigned binary numbers stored at location X and X+1
7. Write a program division of two Binary Numbers stored at location X and X+1. Display the quotient in address field and remainder in data field
8. Write a program to find the smallest of N one byte numbers. Value of N is stored at location X and the numbers from location X+1. Display the smallest number in the data field and its address in the address field
9. Write a program to display a MOD N binary up counter and down counter. Display the count in address/data field. Generate a 0.5 sec delay between the counts. If clock frequency is 3 MHz
10. Configure ports A, B, C in the output mode in 8155 without timer operation and observe the output with the help of LEDs
11. Configure ports A, B, C in the input mode in 8155 with the help of LEDs without timer operation and observe the output.
12. Configure ports A, B, C in the output mode in 8255 without timer operation and observe the output with the help of LEDs
13. Configure ports A, B, C in the input mode in 8255 with the help of LEDs without timer operation and observe the output
14. Configure ports A, B, C in the output mode in 8255 without timer operation and code for getting square signal & observe the output with the help of LEDs
15. Stepper motor Interfacing/Speed Control & Direction Control using 8085 and 8051
16. Write the Embedded C Program and Assembly Program to Display “EMBEDDED LAB” on LCD and compare Output of both program and draw the conclusion
17. Write the Embedded C Program and Assembly Program for Wave Drive Stepper Motor System. And observe the output in the Proteus
18. Write the Embedded C Program and Assembly Program for Half Wave Drive Stepper Motor System. And observe the output in the Proteus.

COURSE OUTCOMES

On completion of the course, student will be able to:

CO1 – Understand the internal architecture of 8085 microprocessor and 8051 microcontroller thoroughly.
CO2 – program assembly language codes for 8085 microprocessor and 8051 microcontroller.
CO3 – apply C language based coding on different problems involving 8051 microcontrollers
CO4 – understand the interfacing of 8085 microprocessor and 8051 microcontroller with different peripherals
CO5 – understand the application of basic concepts of 8085 microprocessor and 8051 microcontroller related to timers, counters, interrupts.
CO6 – engage the students in formulating practical solutions from the theoretical knowledge gained into electrical engineering domain applications

END SEMESTER LABORATORY EXAMINATION PATTERN

Max. Marks: 100

Continuous evaluation	50 marks
End semester examination and Viva-voce	50 marks

20IF201T					Industry 4.0					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
2	0	0	2	2	25	50	25	--	--	100

COURSE OBJECTIVES

1. To interpret the core elements and basic technologies of Industry 4.0
2. To understand how the core elements and technologies of Industry 4.0 are interconnected
3. To develop a holistic approach to improve processes and products with Industry 4.0

UNIT I: INDUSTRY 4.0 – CONCEPTS & TERMINOLOGIES **08 Hrs.**

Industry 4.0, Smart business model, Technology road-map, Sensing & actuation, Communication, Internet of things (IoT), Cyber Physical Systems and Next Generation Sensors, Visualization, Cloud Computing.

UNIT II: SMART WORLD & SUSTAINABLE ENVIRONMENT **08 Hrs.**

Sensors and their integration, Renewable Energy System, Hybrid Energy System, Smart Grid, Smart Metering, Communication Protocols, 5G Technology, Smart Agriculture, Smart Infrastructure, Physiological Sensors, Human Machine Interface.

UNIT III: SMART MANUFACTURING **08 Hrs.**

Automation Systems, Additive Manufacturing, Micro-Electro-Mechanical Systems (MEMS), Smart Factories and Interconnection, Advanced Robotics – Autonomous and Swarm, Self-Propelled Vehicles, Drones–Unmanned Aerial Vehicle (UAV), 3d Printing, Spacecrafts.

UNIT IV: TRANSFORMING TECHNOLOGIES IN BIOENGINEERING **08 Hrs.**

Establishment of Smart Biotechnology Factory, Artificial Intelligence in Bioprocess Technology, 3D Bio Printing for Tissue Engineering, Simulation Tools, RSM and Box Model, Cyber Physical System based Telemedicine, Real Time Biosensors, Bio nanotechnology, biofuel.

Total Hours 32 Hrs.

COURSE OUTCOMES

On completion of the course, student will be able to

CO1 – Understand the core elements and basic technologies for Industry 4.0
CO2 – Apply the different computational techniques and algorithms for realizing Industry 4.0
CO3 – Transform the traditional business approach by integrating the data and intelligence
CO4 – Develop the traditional industries with intelligent and automated machines
CO5 – Utilize data and intelligence for the development of Smart World
CO6 – Understand the concept, significance and means to achieve sustainable development

TEXT/REFERENCE BOOKS

1. Ustundag Alp, and EmreCevikcan, Industry 4.0: Managing the Digital Transformation, Springer,First Edition, 2018
2. Kaushik Kumar, DivyaZindani, and J. Paulo Davim, Digital Manufacturing and Assembly Systems in Industry 4.0., CRC Press, Taylor & Francis First Edition, 2019.
3. Antonella Petrillo, Raffaele Cioffi, and Fabio De Felice, Digital Transformation in Smart Manufacturing., IntechOpen Publisher, First Edition, 2018.
4. J. Ekanayake, K. Liyanage, J. Wu, A. Yokoyama and N. Jenkins, Smart Grid: Technology and Applications, John Wiley and Sons Ltd., First Edition, 2012
5. Alasdair Gilchrist, Industry 4.0: The Industrial Internet of Things, Apress, First Edition, 2016
6. Ibrahim Garbie, Sustainability in Manufacturing Enterprises: Concepts, Analyses and Assessments for Industry 4.0, Springer, First Edition, 2016

END SEMESTER EXAMINATION QUESTION PAPER PATTERN

Max. Marks: 100	**Exam Duration: 3 Hrs**
Part A/Question: 4 Questions, one from each unit, each carrying 15 marks	60 Marks
Part B/Question: 4 Questions, one from each unit, each carrying 10 marks	40 Marks

20IF201P					Industry 4.0 Laboratory					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
0	0	2	1	2	--	--	--	50	50	100

COURSE OBJECTIVES

1. To comprehend the concept and significance of Industry 4.0
2. To understand core elements and technologies of Industry 4.0 through simulation and experimental studies
3. To explore different software packages and hardware elements involved in realization of Industry 4.0

List of Experiments

1. Basic computations using Python programming.
2. Use simulations to understand the performance/behavior of a system by (i) creating a computational environment that mimics the real world, (ii) generating (synthetic) or loading data from sources, and (iii) testing the hypothesis
3. Introduction to MATLAB programming and SIMULINK
4. 3D printing of Airfoil through rapid prototyping 3D printer
5. Dynamic simulation of drone (unmanned air vehicle) through MATLAB/SIMULINK
6. ANSYS simulation of bending of a beam in an earthquake resist-building
7. Introduction to Arduino Embedded platform.
8. Design of line follower autonomous vehicle.
9. Design of smart meter for recording the electricity consumption
10. Design of smart lighting with the help of proximity sensors.

COURSE OUTCOMES

On completion of the course, student will be able to

CO1 – Understand the concept of Industry 4.0 and its significance
CO2 – Understand the resource requirements for the implementation of Industry 4.0
CO3 – Learn the Simulation Packages for Industry 4.0
CO4 – Explore the concept of Smart Infrastructure through simulation studies
CO5 – Inspect embedded platform applications for Industry 4.0
CO6 – Synthesise the solution for the given Industry 4.0 related problem

END SEMESTER LABORATORY EXAMINATION PATTERN

Max. Marks: 100

Continuous evaluation	50 marks
End semester examination and Viva-voce	50 marks

TP210					Industrial Orientation					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
-	-	-	2	--	--	--	--	--	100	100

COURSE OBJECTIVES

1. To introduce students to the working environment of the industry
2. To understand the different departments involved in an industry for developing a product or offering a service
3. To learn about the significance of the theoretical knowledge being imparted in the lecture sessions in working of an industry

Students are required to visit industries to observe the working of an industry. The students will have the opportunity to understand various industrial processes, departments involved and manufacturing process. Moreover, the students will get an opportunity to see the theoretical knowledge acquired being put into practice in industry.

COURSE OUTCOMES

On completion of the course, student will be able to:

CO1 – Understand the working of industry
CO2 – Acquire knowledge about the different departments involved in the functioning of an industry
CO3 – Learn about the products being developed or services being offered by the industry
CO4 – Understand the safety procedures followed by industry while working on shop floor
CO5 – Learn how the theoretical knowledge is utilized for product development or for services being offered
CO6 – Acquire the necessary skills for technical report writing

END SEMESTER EXAMINATION PATTERN

Max. Marks: 100

End semester examination and Viva-voce 100 marks

20EE209T					Energy and Water Nexus					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
3	-	-	3	3	25	50	25	--	--	100

COURSE OBJECTIVES

1. To understand the significance of water and energy efficiency.
2. To gather knowledge about interdependency of energy and water use.
3. To identify the processes requiring energy for water use and water for energy generation.

UNIT I:INTRODUCTION TO ENERGY-WATER NEXUS **08 Hrs**

What is Energy-Water nexus?, Energy-Water flow diagram, Unsustainable and sustainable development, Energy and water challenges, Need for integrating energy and water, Population, Hunger and Millennium Development Goals (MDGs), UN sustainable development goals (SDGs), Climate change and its impact, Electricity consumption in the water sector by process, Societal aspects: rural livelihood scenario, urban scenario,

UNIT II:ENERGY FOR WATER: ENERGY USE FOR WATER DEMAND **10 Hrs**

Energy and its measurement, Drivers of water demand, Characteristics and Properties of Water: Abundance, Thermal Properties, Phase Transitions, Other Properties, Regional and Temporal Variability in Water Accessibility, Linkages between the Fuels Life Cycle and Water Quality. Energy consumption for drinking water supply and wastewater treatments, Energy for ground-water pumping, Energy for desalination, Energy pricing and costs

UNIT III:WATER FOR ENERGY: WATER USE FOR ELECTRICITY GENERATION **13 Hrs**

Drivers of energy demand, Sectoral energy demand: domestic, commercial, industrial, agricultural
fossil-fuel based power generation processes: Coal fuel cycle, Thermoelectric Cooling, Oil and Natural gas, Nuclear power, renewable power sources: Geothermal, Hydroelectricity, Solar thermal power generation, solar photovoltaic, wind energy Transportation Fuels Production: Water Life Cycle Management in Fossil Fuels Production, Water Consumption Intensity for Fuels Production, Water Pricing and Costs

UNIT IV: CASE STUDIES AND FUTURISTIC INNOVATIONS **09 Hrs**

Case studies, Circular economy, Challenges and opportunities, Data, Modelling, and Analysis, thinking to enhance energy-water nexus thinking, Sector-Specific Water-Energy Landscape for Decision Making, Challenges and opportunities

TOTAL HOURS 40 Hrs

COURSE OUTCOMES

On completion of the course, student will be able to:

- CO1 - Examine the concept of Water and Energy usage.
- CO2 - Analyse behaviour of energy consumption for water demand.
- CO3 - Analyse behaviour of water usage for energy demand.
- CO4 - Understand the concept of interlinkage of Energy and Water.
- CO5 - Gather knowledge through national case studies.
- CO6 - Gather knowledge through international case studies.

TEXT/REFERENCE BOOKS

1. Ernest J. Moniz (2014), "**The Water-Energy Nexus: Challenges and Opportunities**", U.S. Department of Energy.
2. Gleick, P. H. (1994). "**Water and Energy**." Annual Review of Energy and the Environment, 19(1), 267–299.
3. Khatavkar, P., & Mays, L. W. (2017 a) "**Model for the Real-Time Operation of Water Distribution Systems under Limited Power Availability**". In World Environmental and Water Resources Congress 2017 (pp. 171–183).
4. Khatavkar, P., & Mays, L. W. (2017). Testing an Optimization/Simulation Model for the Real-Time Operations of Water Distribution Systems under Limited Power Availability. In Congress on Technical Advancement 2017 (pp. 1–9).

END SEMESTER EXAMINATION QUESTION PAPER PATTERN

Max. Marks: 100	**Exam Duration: 3 Hrs**
Part A/Question: 4 Questions, one from each unit, each carrying 20 marks	80 Marks
Part B/Question: 2 Questions, one each from unit II and III, each carrying 10 marks	20 Marks